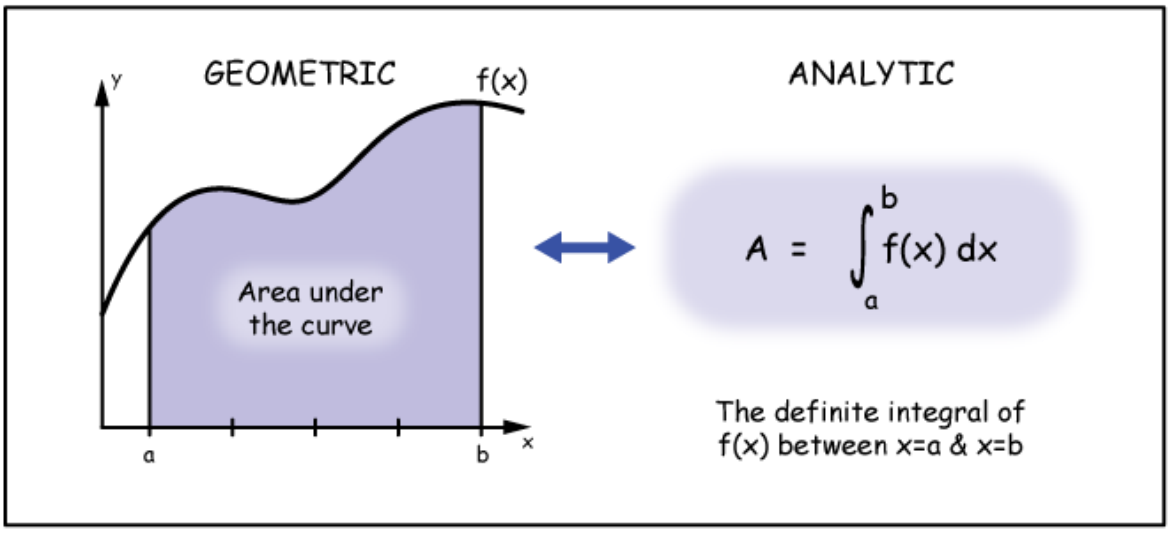
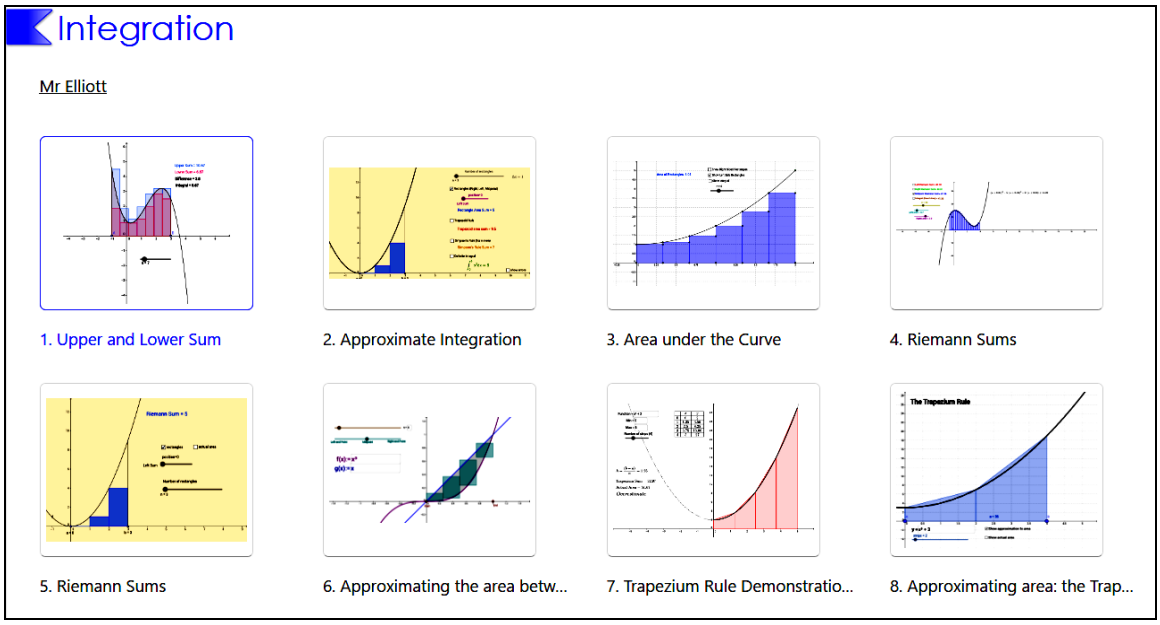
Area under curves（曲線下的面積）



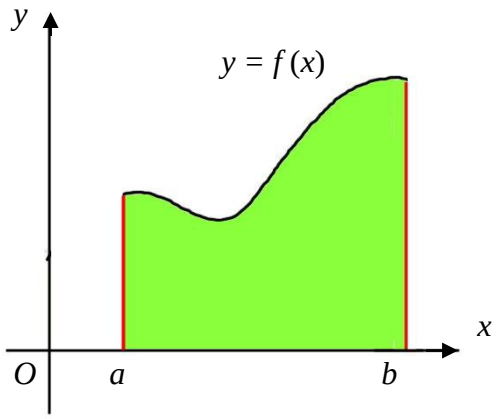
Students who are interested in knowing more about Definite Integration for Areas under Curve can refer to the following GeoGebra Book:

https://www.geogebra.org/b/NbMfChDF



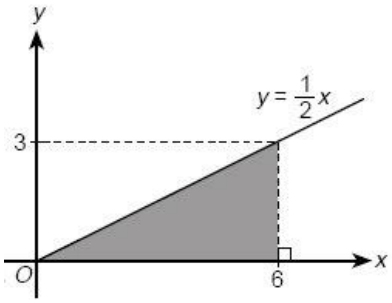
Application of Integration : Area under curves

1. Area between one curve and *x*-axis

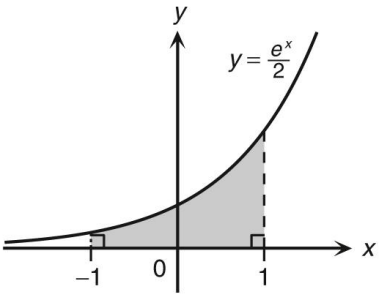


$$\text{Area} = \int_a^b f(x)\,dx$$

e.g.1. Find the shaded area in the figure.



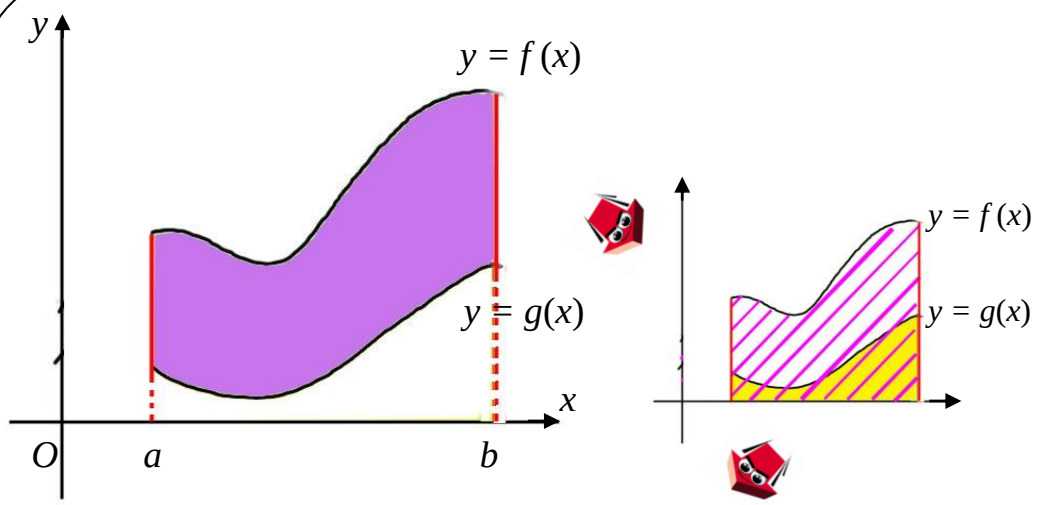
e.g.2 Find the area of the shaded region in the figure.



Not for sale

Application of Integration : Area under curves

2. Area between two curves – Type I

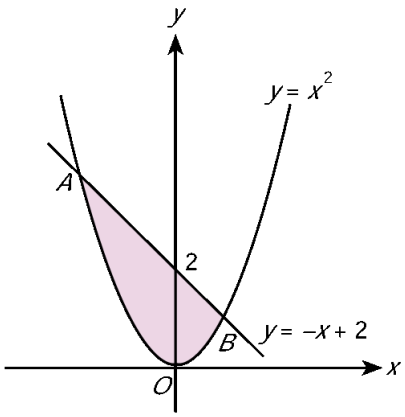


$$\text{Area} = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

$$= \int_a^b [\ f(x) - g(x)\ ]\,dx$$

upper *lower*

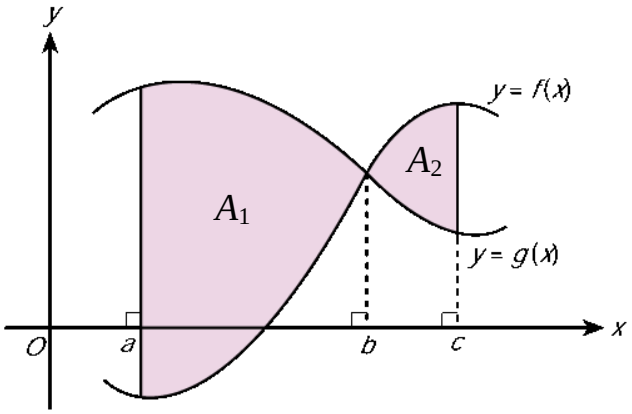
e.g.3 Find the shaded area in the figure.



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Application of Integration : Area under curves

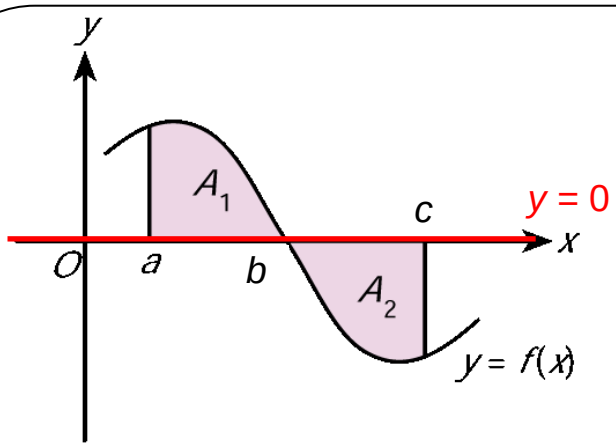
3. Area between two curves – Type II



若兩條曲線是相交的，則我們要將面積分為兩部份計算...

$$\text{Area} = \overbrace{\int_a^b [g(x) - f(x)]\,dx}^{\text{Area } A_1} + \overbrace{\int_b^c [f(x) - g(x)]\,dx}^{\text{Area } A_2}$$

upper lower upper lower



若曲線與 **X** 軸是相交的，則我們亦要將面積分為兩部份計算...

$$\text{Area} = \overbrace{\int_a^b [f(x) - 0]\,dx}^{\text{Area } A_1} + \overbrace{\int_b^c [0 - f(x)]\,dx}^{\text{Area } A_2}$$

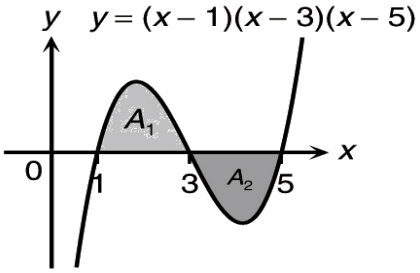
upper lower upper lower

$$= \int_a^b f(x)\,dx - \int_b^c f(x)\,dx$$

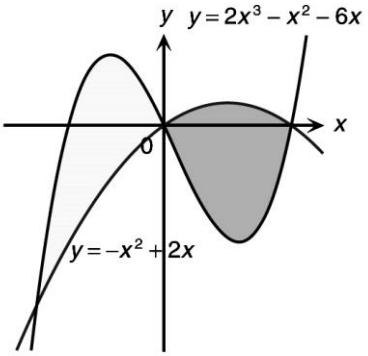
Not for sale

Application of Integration : Area under curves

e.g.4 The figure shows the graph of $y = (x - 1)(x - 3)(x - 5)$.
Find the shaded areas A_1 and A_2 respectively.



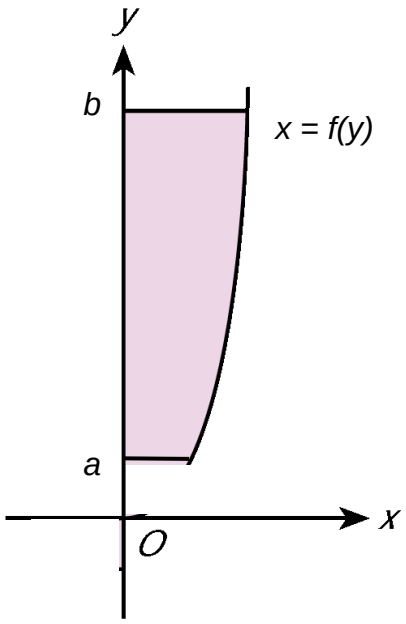
e.g.5 Find the area of the shaded region in the figure.



Not for sale

Application of Integration : Area under curves

4. Area between one curve and *y*-axis



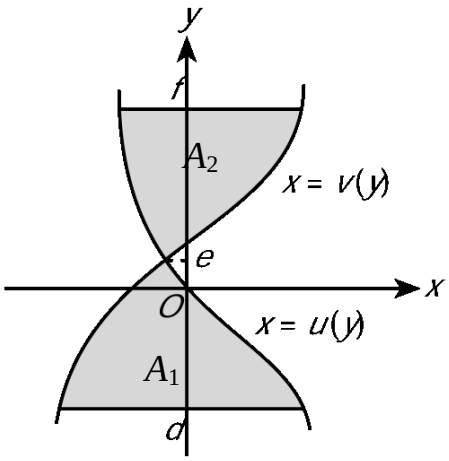
若曲線是以 ***y*** 表示 **(in terms of *y*)**

e.g. $x = y^3 + 2y + 1$

則我們要用 **y** 做主項進行積分**!**

$$\text{Area} = \int_a^b f(y)\,dy$$

5. Area between two curves – Type III



若曲線與 ***y*** 軸是相交的，則我們亦要將面積分為兩部份計算**...**

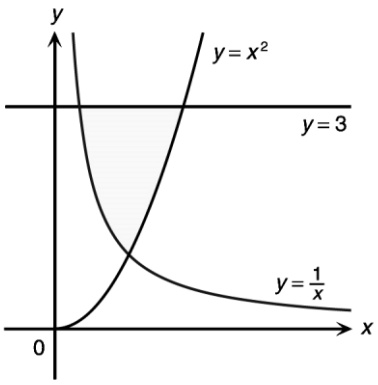
$$\text{Area} = \overbrace{\int_d^e [u(y) - v(y)]\,dy}^{\text{Area } A_1} + \overbrace{\int_e^f [v(y) - u(y)]\,dy}^{\text{Area } A_2}$$

right *left* *right* *left*

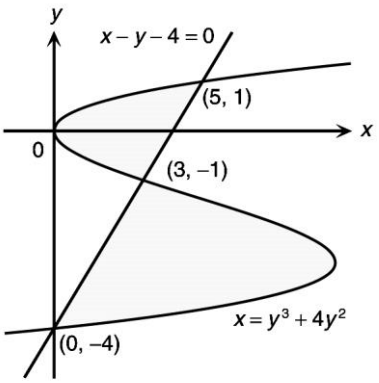
Not for sale

Application of Integration : Area under curves

e.g.6 Find the area of the shaded region in the figure.



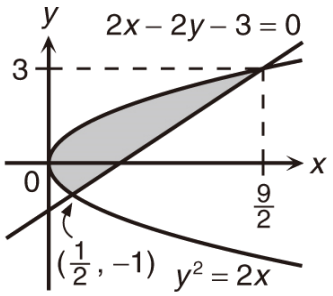
e.g.7 Find the area of the shaded region in the figure.



Not for sale

Application of Integration : Area under curves

e.g.8 The figure shows the graphs $2x - 2y - 3 = 0$ and $y^2 = 2x$.



Find the area of the shaded region in the figure by integrating

(a) with respect to x,

(b) with respect to y.

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HKCEE Questions (Level 1)

1. ***[A.Math 1992 Paper II Q.6]***

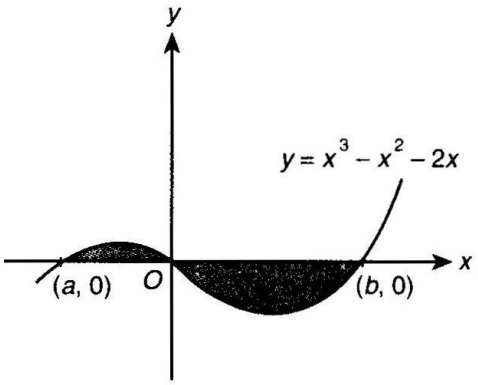


Fig. 6.2

The curve $y = x^3 - x^2 - 2x$ cuts the x-axis at the origin and the points $(a, 0)$ and $(b, 0)$, as shown in Fig. 6.2.

(a) Find the values of a and b.

(b) Find the total area of the shaded parts. ***(6 marks)***

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HKCEE Questions (Level 1)

2. ***[A.Math 1994 Paper II Q.7]***

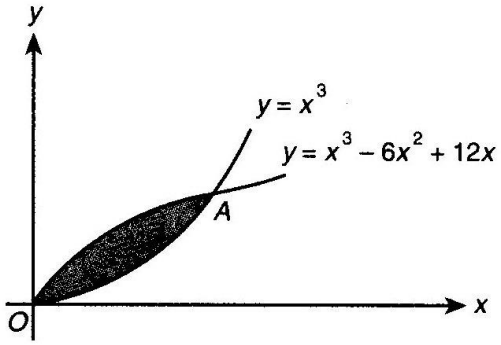


Fig. 6.4

Fig. 6.4 shows two curves $y = x^3$ and $y = x^3 - 6x^2 + 12x$ intersecting at the origin and a point *A*.

(a) Find the coordinates of *A*.

(b) Find the area of the shaded region in Fig 6.4. ***(6 marks)***

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HKCEE Questions (Level 1)

3. ***[A.Math 1988 Paper II Q.3]***

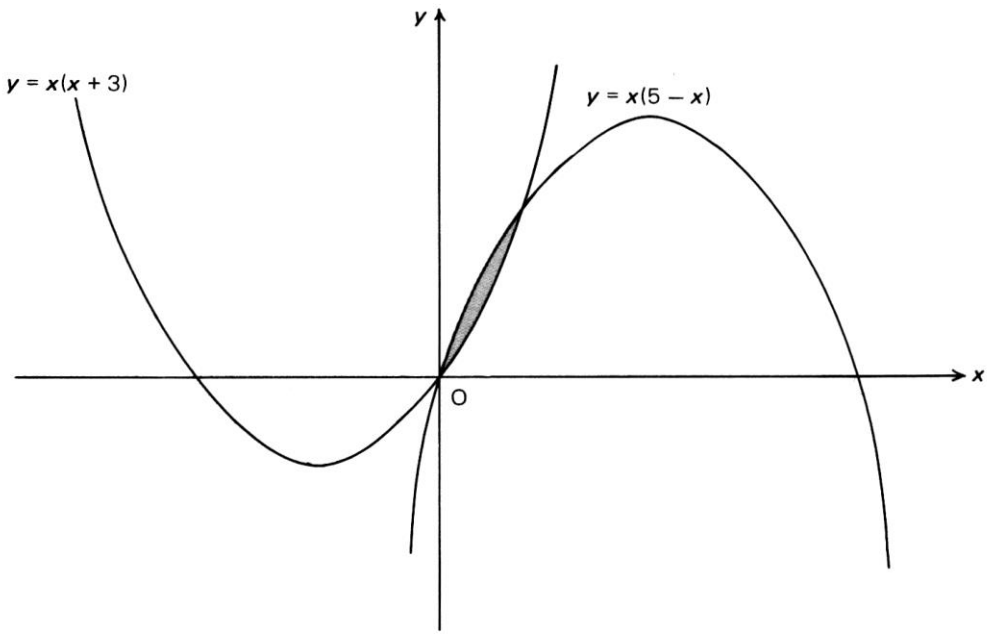


Fig. 7

In Fig.7, the shaded region is bounded by the two curves and $y = x(x+3)$ and $y = x(5-x)$. Find the area of the shaded region. ***(5 marks)***

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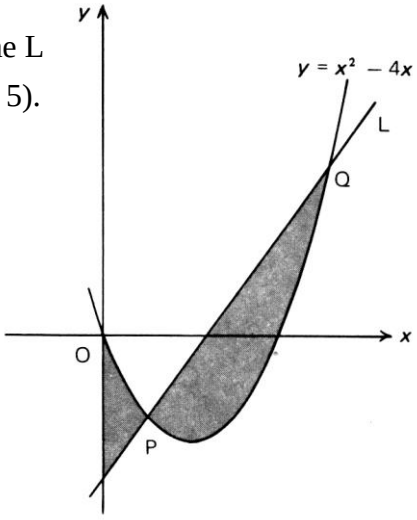
HKCEE Questions (Level 1)

4. ***[A.Math 1983 Paper II Q.4]***

Fig. 5 shows the curve $y = x^2 - 4x$. A straight line L Intersects the curve at the point P(1, –3) and Q(5, 5). Find

(a) The equation of L, and

(b) The area of the shaded region.



(6 marks)

Not for sale

HKCEE Questions (Level 1)

5. ***[A.Math 2008 Q.10]***

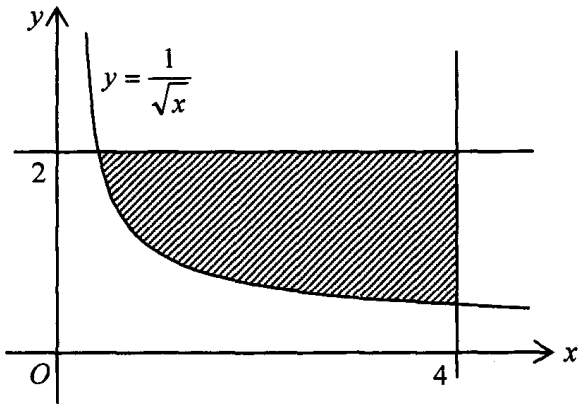


Figure 2

Find the area of the shaded region bounded by the curve $y = \frac{1}{\sqrt{x}}$ and the straight lines $y = 2$, $x = 4$ (see Figure 2). ***(5 marks)***

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HKCEE Questions (Level 2)

6. ***[A.Math 2002 Q.6]***

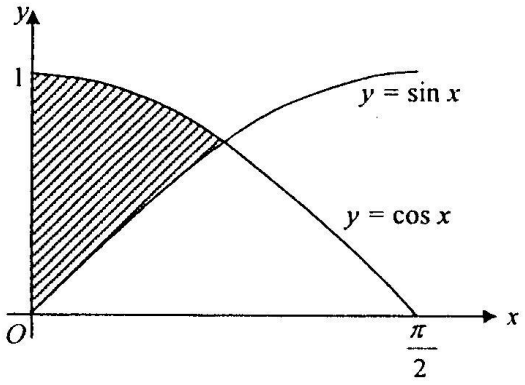


Figure 1

Figure 1 shows the curves $y = \sin x$ and $y = \cos x$. Find the area of the shaded region. ***(5 marks)***

Not for sale

HKCEE Questions (level 2)

7. ***[A.Math 2009 Q.10]***

(a) Solve $8\cos x = \sec^2 x$ for $0 < x < \frac{\pi}{2}$.

(b)

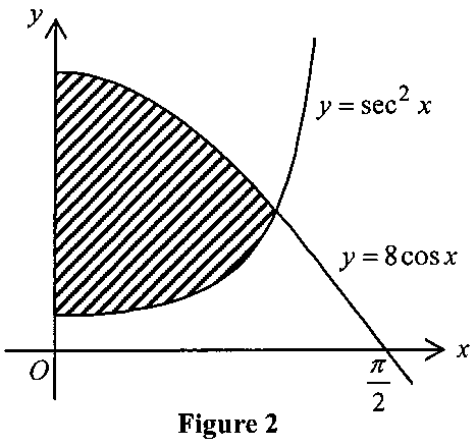


Figure 2

Figure 2 shows the graphs of $y = 8\cos x$ and $y = \sec^2 x$.
Find the area of the shaded region. ***(5 marks)***

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HKCEE Questions (level 2)

8. ***[A.Math 2007 Q.6]***

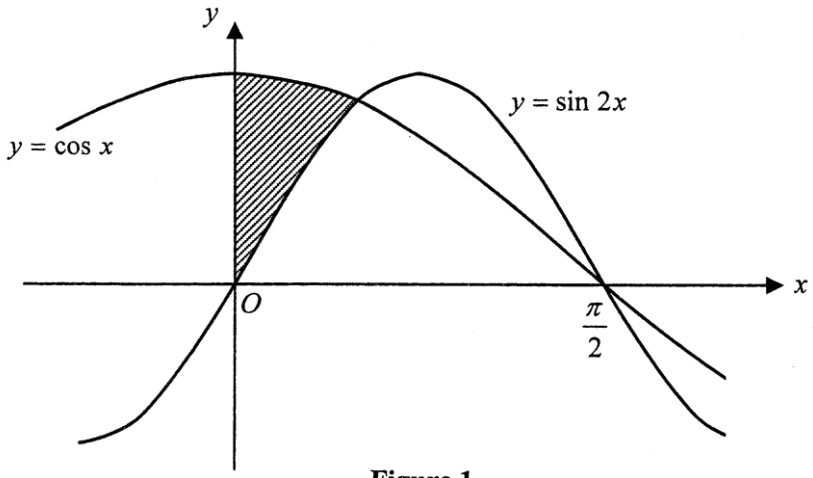


Figure 1

Figure 1 shows the graphs of $y = \sin 2x$ and $y = \cos x$. Find the area of shaded region. ***(5 marks)***

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HKCEE Questions (level 2)

9. ***[A.Math 2005 Q.13]***

(a) Find $\int \sin \pi x dx$.

(b)

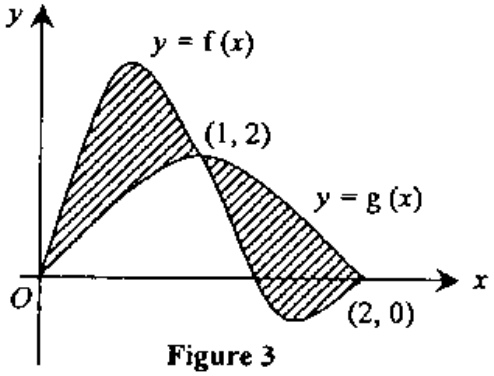


Figure 3

Figure 3 shows two curves $y = f(x)$ and $y = g(x)$ intersecting at three points (0, 0), (1, 2) and (2, 0) for $0 \le x \le 2$. It is given that $f(x) - g(x) = 2\sin \pi x$. Find the area of the shaded region as shown in Figure 3. ***(6 marks)***

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HKCEE Questions (level 2)

10. ***[M&S 2000 Q.3]***

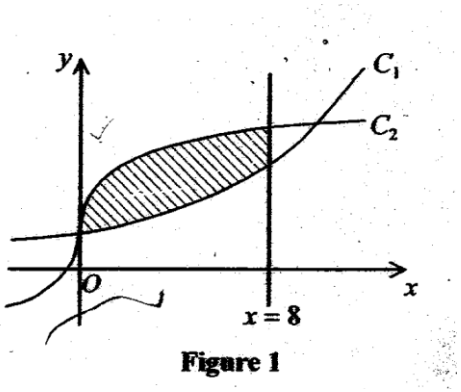


Figure 1

Figure 1 shows the graphs of the two curves $C_1 : y = e^{\frac{x}{8}}$ and $C_2 : y = 1 + x^{\frac{1}{3}}$.

Find the area of the shaded region. ***(5 marks)***

Not for sale

HKCEE Questions (Level 2)

11. ***[A.Math 2010 Q.3]***

Let $I = \int_{-2}^{1} x(1-x)(x+2)\,dx$.

(a) Find the value of *I*.

(b)

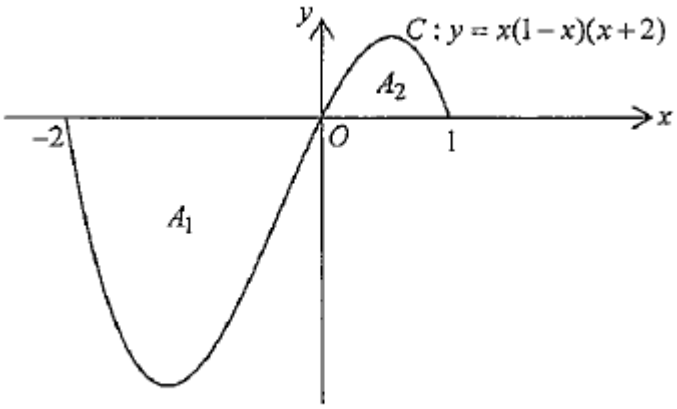


Figure 1

Figure 1 shows the graph of $C: y = x(1-x)(x+2)$ for $-2 \le x \le 1$. Let A_1 denote the area of the region bounded by *C* and the *x*-axis when $x < 0$, and A_2 denote the area of the region bounded by C and the x-axis when $x \ge 0$. Without finding the values of A_1 and A_2, express *I* in terms of A_1 and A_2. ***(4 marks)***

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HKCEE Questions (Level 2)

12. ***[A.Math 1996 Paper II Q.5]***

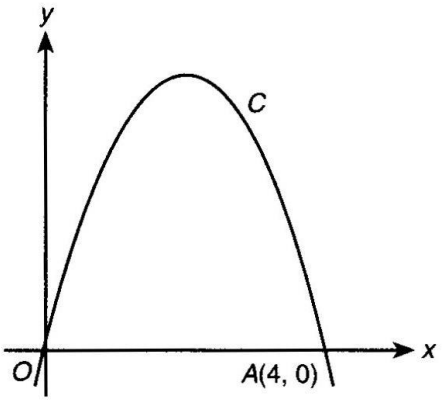


Fig. 6.6 (a)

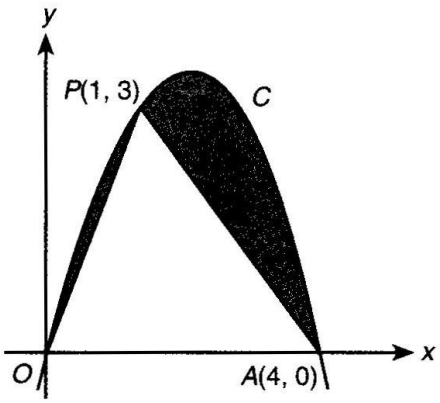


Fig. 6.6 (b)

The curve C: $y = 4x - x^2$ cuts the x-axis at the origin and the point $A(4, 0)$ as shown in Fig. 6.6 (a).

(a) Find the area of the region bounded by C and the line segment OA.

(b) In Fig 6.6 (b), the shaded region is enclosed by the curve C and the line segments OP and PA, where P is the point (1, 3). Using **(a)**, find the total area of the shaded region. ***(6 marks)***

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HKCEE Questions (Level 2)

13. ***[A.Math 2003 Q.9]***

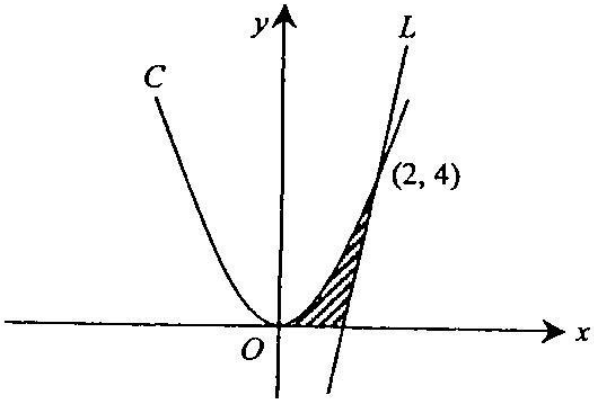


Figure 2

In Figure 2, the curve $C : y = x^2$ and line $L : y = 4x - 4$ intersects at the point (2, 4). Find the area of the shaded region bounded by C, L and the x-axis.

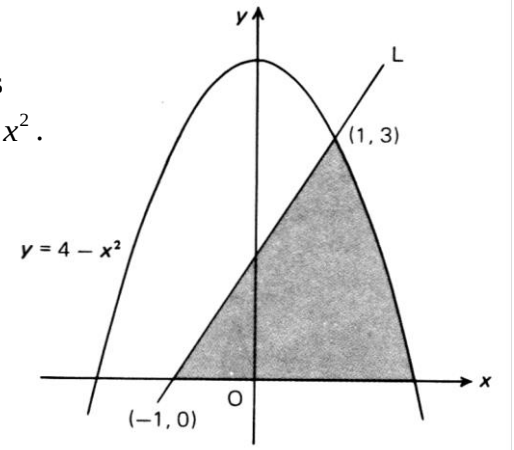
(5 marks)

Not for sale

HKCEE Questions (level 2)

14. ***[A.Math 1985 Paper II Q.4]***

In Fig. 6, the straight line L uts the *x*-axis at the point (–1, 0) and the curve $y = 4 - x^2$. Find the area of the shaded part.



(5 marks)

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HKCEE Questions (level 2)

15. ***[A.Math 1999 Paper II Q.4]***

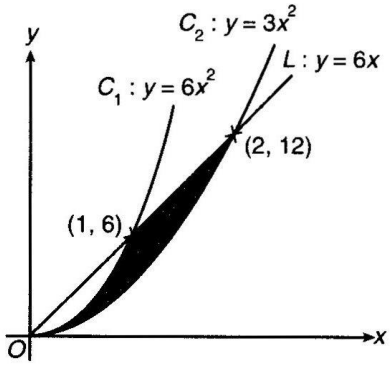


Fig. 6.8

In Fig. 6.8, the line L: $y = 6x$ and the curves C_1: $y = 6x^2$ and C_2: $y = 3x^2$ all passes through the origin. L also intersects C_1 and C_2 at the points (1, 6) and (2, 12) respectively. Find the area of the shaded region. ***(5 marks)***

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HKDSE Questions (level 2)

16. ***[HKDSE M1 Practice Paper Q.5]***

Consider two curves $C_1 : y = 1 - \frac{e}{e^x}$ and $C_2 : y = e^x - e$.

(a) Find the x-coordinates of all the points of interscetion of C_1 and C_2.

(b) Find the area of the region bounded by C_1 and C_2.

(5 marks)

Not for sale

HKDSE Questions (level 2)

17. ***[HKDSE M1 2012 Q.5]***

The slope of the tangent to a curve S at any point (x, y) on S is given by $\frac{dy}{dx} = e^{2x}$.

Let L be the tangent to S at the point $A(0, 1)$ on S.

(a) Find the equation of S.

(b) Find the equation of L.

(c) Find the area of the region bounded by S, L and the line $x = 1$.

(7 marks)

Not for sale

HKDSE Questions (Level 3)

18. ***[HKDSE M1 2013 Q.3]***

Consider the curve $C: y = x(x-2)^{\frac{1}{3}}$ and the straight line L that passes through the origin and is parallel to the tangent to C at $x = 3$.

(a) Find the equation of L.

(b) Find the x-coordinates of the two intersecting points C and L.

(c) Find the area bounded by C and L.

[Note: For definite integrals, answer obtained by using numerical integration functions in calculators are not accepted.]

(8 marks)

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HKASL Questions (Level 3)

19. ***[M&S 1995 Q.6]***

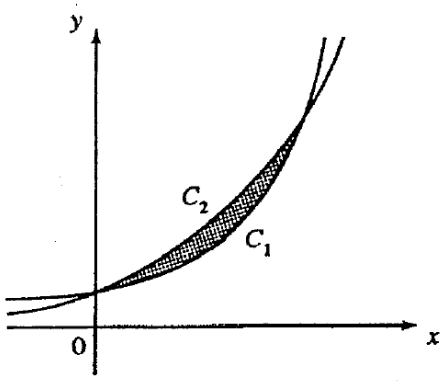


Figure 1

Figure 1 shows graphs of the two curves $C_1 : y = 2^{2x} + 4$ and $C_2 : y = 5(2^x)$.

(a) Find the coordinates of the points of intersection of C_1 and C_2.

(b) If $2^x = e^{ax}$ for all x, find the value of a.
Hence, or otherwise, find the area of the shaded region in Figure 1 bounded by C_1 and C_2.

(8 marks)

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HKASL Questions (Level 3)

20. ***[M&S 1996 Q.4]***

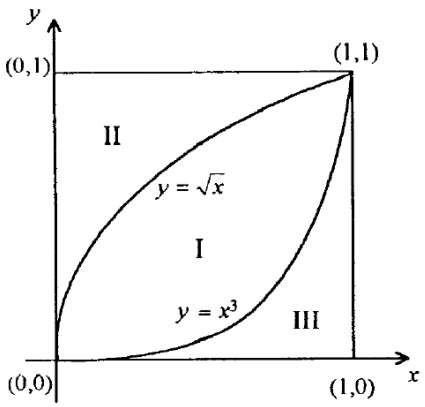


Figure 1

Figure 1 shows a unit square target for shooting on the rectangular coordinate plane. The target is divided into three regions Ⅰ, Ⅱ and Ⅲ by the curves $y=\sqrt{x}$ and $y=x^3$. The scores for hitting the regions Ⅰ, Ⅱ and Ⅲ are 10, 20 and 30 points respectively.

(a) Find the areas of the three regions.

(b) Suppose a child shoots randomly at the target twice and both shots hit the target. Find the probability that he will score 40 points.

(7 marks)

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HKDSE Questions (level 3)

21. ***[HKDSE 2014 Q.6]***

. (a) Find $\int xe^{-x}dx$.

(b)

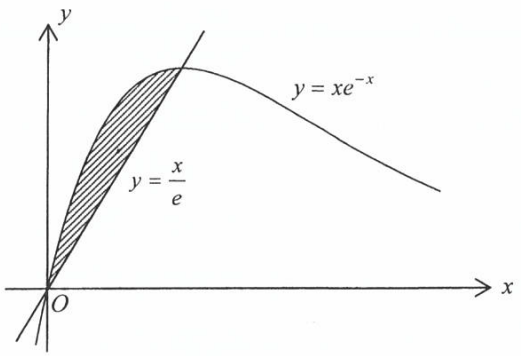


Figure 1

Figure 1 shows the shaded region bounded by the curve $y = xe^{-x}$ and the straight line $y = \frac{x}{e}$. Find the area of the shaded region.

(6 marks)

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HKDSE Questions (level 3)

22. ***[HKDSE 2012 Q.14]***

Consider the curve $\Gamma : y = kx^p$, where $k > 0$, $p > 0$. In Figure 7, the tangent to Γ at $A(a, ka^p)$ cuts the x-axis $B(-a,0)$, where $a > 0$.

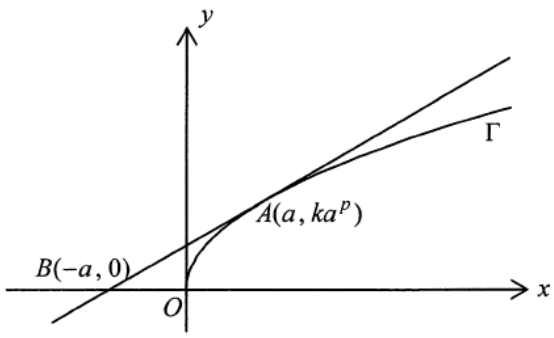


Figure 7

(a) Show that $p = \frac{1}{2}$. ***(3 marks)***

(b) Suppose that $a = 1$. As shown in Figure 8, the circle C, with radius 2 and centre on the y-axis, touches Γ at point A.

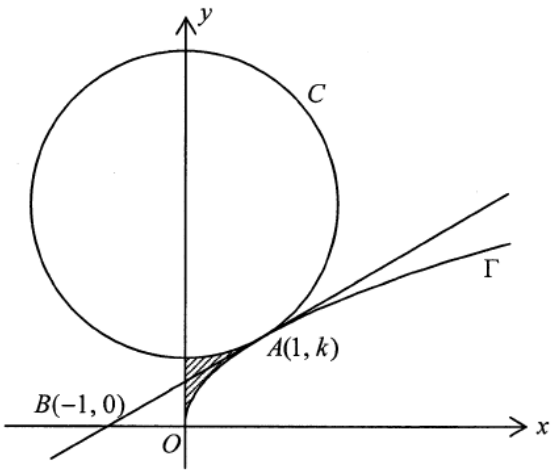


Figure 8

(i) Show that $k = \frac{2\sqrt{3}}{3}$.

(ii) Find the area of the shaded region bounded by Γ, C and the y-axis. ***(9 marks)***

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23. ***[HKDSE 2015 Q.9]***

Define $f(x) = \dfrac{x^2+12}{x-2}$ for all $x \neq 2$.

(a) Find $f'(x)$. ***(2 marks)***

(b) Prove that the maximum value and the minimum value of $f(x)$ are -4 and 12 respectively. ***(4 marks)***

(c) Find the asymptote(s) of the graph of $y = f(x)$. ***(3 marks)***

(d) Find the area of the region bounded by the graph of $y = f(x)$ and the horizontal line $y = 14$. ***(4 marks)***

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HKCEE Questions (level 3)

24. ***[A.Math 2006 Q.13]***

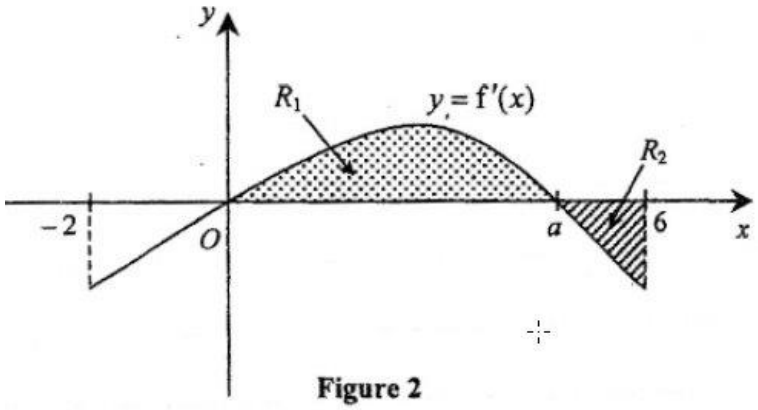


Figure 2

Let $f(x)$ be a polynomial. Figure 2 shows a sketch of the curve $y = f'(x)$, where $-2 \le x \le 6$. The curve cuts the x-axis at the origin and (a, 0), where $0 < a < 6$. It is known that the areas of the shaded regions R_1 and R_2 as shown in Figure 2 are 3 and 1 respectively.

(a) Write down the x-coordinates of the maximum and minimum points of the curve $y = f(x)$ for $-2 < x < 6$.

(b) It is known that $f(-2) = 2$ and $f(0) = 1$.

(i) By considering $\int_0^a f'(x)\,dx$, find the value of $f(a)$.

(ii) In Figure 3, sketch the curve $y = f(x)$ for $-2 \le x \le 6$.

(7 marks)

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HKCEE Questions (level 3)

24. ***[A.Math 2006 Q.13] (Cont'd)***

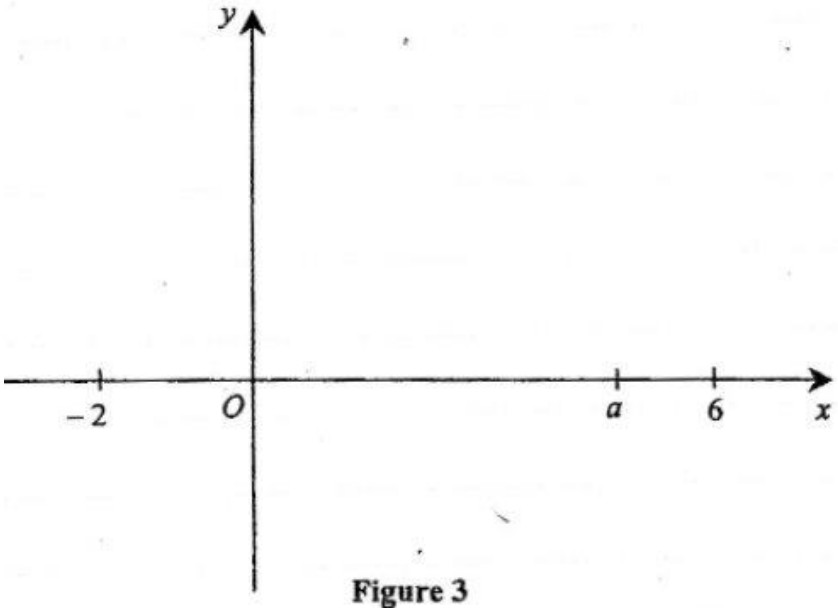


Figure 3

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HKCEE Questions (Level 3)

25. ***[A.Math 2004 Q.17 (b)]***

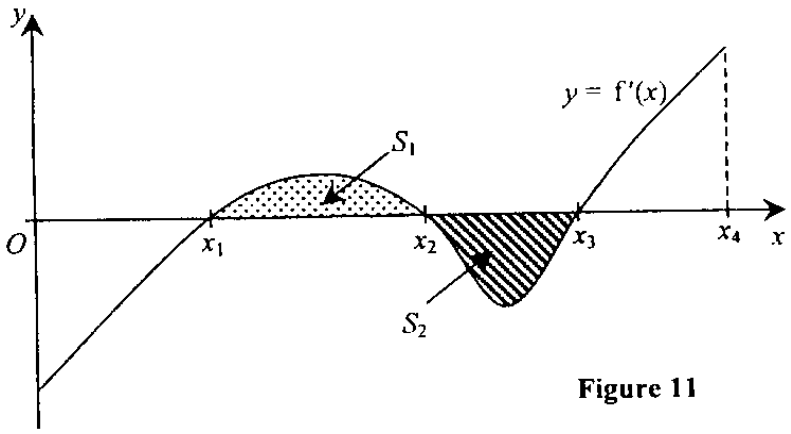


Figure 11

Let $f(x)$ be a continuous function. Figure 11 shows a sketch of the graph of $y = f'(x)$ for $0 \le x \le x_4$. It is known that the areas of the shaded regions S_1 and S_2 as shown in Figure 11 are equal.

(i) Show that $f(x_1) = f(x_3)$.

(ii) Furthermore, $f(0) = f(x_4) = 0$ and $f(x) \neq 0$ for $0 \le x \le x_4$. In Figure 12, draw a sketch of the graph of $y = f(x)$ for $0 \le x \le x_4$.

(5 marks)

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HKCEE Questions (level 3)

25. ***[A.Math 2004 Q.17 (b)] (Cont'd)***

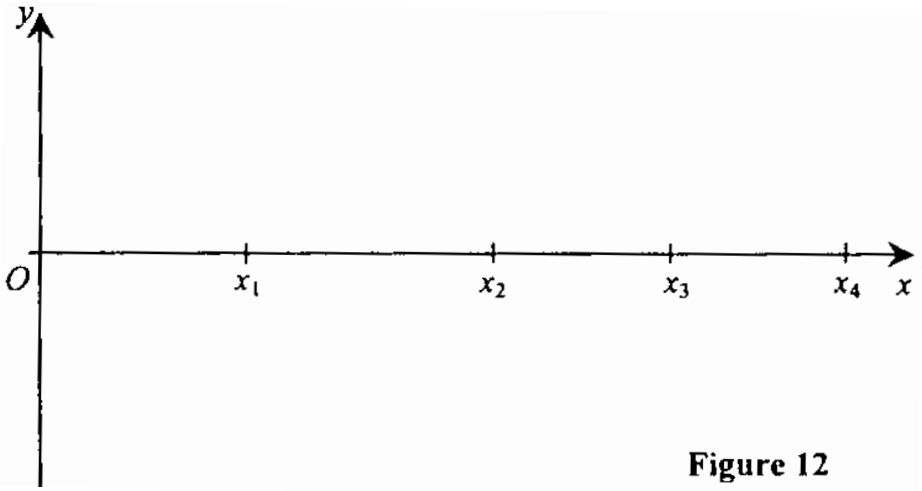


Figure 12

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